Predicting Molecular subtypes of Endometrial Cancer from Clinical Features

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ABSTRACT

Endometrial cancer is a disease which requires specific care based on its tumor subtypes and certain subtypes can be completely cured when found early. These subtypes are usually identified through histopathology and genomic sequencing which are costly and time consuming. The aim of this project is to apply machine learning algorithms to predict the three molecular subtypes of endometrial cancer (Cancer Type Detailed) using demographic and pre-treatment clinical features from The Cancer Genome Atlas (TCGA) dataset. This project evaluates six algorithms: Logistic Regression, Decision Trees, Random Forests, Naïve Bayes, Neural networks and XGBoost. Synthetic Minority Oversampling Technique (SMOTE) was applied to handle severe class imbalance and Recursive Feature Elimination (RFE) for feature selection. It is evident from the experiments that a parsimonious decision tree model with 25 features achieved optimal balance with accuracy of 76.4% and F1-macro of 54.48%, outperforming the more complex models. Integrating this model into clinical decision support system could help practitioners identify endometrial cancer subtypes at an early stage, giving ample time to plan for treatment and further procedures.

1. INTRODUCTION

Endometrial cancer is one of the types of gynecologic cancers in women. It is a disease in which the healthy cells of the tissue in the endometrium (the inner lining of uterus) are invaded and mutated, which further grow in large numbers to form a malignant tumor. There are 3 different molecular subtypes of endometrial cancer, which are, Uterine Endometrioid Carcinoma, Uterine Serous Carcinoma and Endometrial Carcinoma. These subtypes carry different levels of risk, where Uterine Endometrioid Carcinoma is less aggressive than the other two subtypes and thus the treatment plan varies accordingly. Early identification is critical because detecting a less aggressive subtype on time could help patient get appropriate treatment and help prevent progression to a much more severe subtype.

There is a substantial amount of clinical data that can be collected without the need for invasive procedures which often most people fear. If a system could identify the likely subtype of the disease using non-invasive clinical features, then the practitioner could make more informed decisions about whether to order further invasive procedures to confirm the presence of Carcinoma. Therefore, the research question this project addresses is: Can demographic, histopathological, pre-treatment clinical features accurately predict the molecular subtypes of endometrial cancer?

Most existing works mainly focus on laboratory and genomic data that are available only after the disease has been identified. Some approaches also leverage deep learning models to classify molecular subtypes from histopathological images, but these methods are computationally intensive. Even though these methods are accurate, they rely on information from clinical resources and infrastructure that may not be widely available, especially in low-resource constrained healthcare settings.

This project addresses this challenge by developing an interpretable and parsimonious machine learning approach for predicting molecular subtypes by using readily available clinical and demographic features. In summary, this project makes the following contributions:

- A robust data preprocessing pipeline that handles high dimensionality in categorical features and manages missing data using statistical imputation.
- Address the severe class imbalance inherent in medical datasets using SMOTE to improve the detection of rare subtypes.
- Identify the most predictive clinical biomarkers using RFE, reducing the feature space while maintaining model accuracy.

2. RELATED WORK

There were two review papers on using machine learning for prediction and prognosis of endometrial cancer [1][5]. From these reviews, it is clear that they treat this as a binary classification problem. They also showed that simple logistic regression performed well on their datasets. However, this project approaches the problem as a multi-class classification task to identify histological subtypes using clinical variables.

One paper applied deep learning techniques to classify histologic grading, which again is a binary classification problem and also requires processing pathology images, making it computationally intensive [4]. In contrast, this project uses tabular clinical data that supports lightweight models which can be easily deployed into existing clinical decision-support systems.

A recent study used 36 serological markers along with clinical features, but obtaining such serological marker data can be difficult in many clinical settings [2].

3. A MOTIVATING EXAMPLE

Consider a 54-year-old female patient presenting with post-menopausal bleeding at a community clinic. Her medical records show no prior cancer occurrence, no history of diabetes, hypertension, or colorectal cancer, past use of hormonal contraceptives, four full-term pregnancies, and a body mass index of 32. The practitioner could input these details into the proposed machine learning system integrated within their clinical decision support tool to obtain the probability distribution of the three molecular subtypes. Based on the predicted outcome, the practitioner can then decide whether further invasive procedures are necessary. This helps in identifying the disease early and

may also help avoid unnecessary invasive procedures. The system does not replace pathological diagnosis but assists in timely decision-making and treatment planning.

4. Approach

4.1 Data Collection and Preprocessing

The dataset was obtained from the TCGA Uterine Corpus Endometrial Carcinoma (UCEC) study via cBioPortal [3]. It contains 549 samples with 75 clinical attributes. Out of the 75 attributes, around 51 were post-operative or laboratory-based features that cannot be included, as this project focuses only on demographic and pre-treatment clinical features. Below are the detailed explanations of the remaining 24 variables used:

Feature	Description	Data Type
Diagnosis Age	Age of the patient at the time of diagnosis	Numeric
Cancer Type Detailed	The specific molecular subtype of endometrial cancer	Categorical
JCCC Stage	Clinical cancer stage based on the tumor's extent and spread	Categorical
History Diabetes	Indicates whether the patient has a history of diabetes	Categorical
Ethnicity	Ethnic background of the patient	Categorical
Fraction Genome Altered	Percentage of the genome with copy number alterations	Numeric
Neoplasm Histologic Grade	Degree of abnormality of cancer cells (tumor grade)	Categorical
Patient Height	Height of the patient	Numeric
History colorectal cancer	Whether the patient has previously had colorectal cancer	Categorical
History hormonal contraceptives use	Past use of hormonal contraceptives	Categorical
History menopausal hormone therapy	Past use of hormone therapy after menopause	Categorical
Prior Cancer Diagnosis Occurrence	Indicates if the patient had any prior cancer diagnosis	Categorical
History tamoxifen use	Whether the patient has taken tamoxifen in the past	Categorical
Hypertension diagnosis	Indicates if the patient has been diagnosed with high blood pressure	Categorical
Menopause Status	Whether the patient is pre-menopausal, peri-menopausal, or post-menopausal	Categorical
Mutation Count	Total number of detected gene mutations in the tumor	Numeric

Peritoneal washing	Indicates whether peritoneal washings tested positive or negative	Categorical
Pregncies full term count	Number of full-term pregnancies the patient has had	Categorical
Race Category	Race of the patient	Categorical
Surgical Margin Resection Status	Whether cancer cells were found at the surgical margins	Categorical
Surgical approach at diagnosis	Type of surgery performed at diagnosis (open, laparoscopic)	Categorical
TMB (nonsynonymous)	Tumor mutation burden counting nonsynonymous mutations	Numeric
Tumor invasion percent	Percentage of the uterine wall invaded by the tumor	Numeric
Patient Weight	Weight of the patient	Numeric

The target variable is "Cancer Type Detailed" which is categorical and has 3 classes, which are

- Uterine Endometrioid Carcinoma (UEC) (410 samples – 74.7%)
- Uterine Serous Carcinoma/Uterine Papillary Serous carcinoma (USC/UPSC) (115 samples – 20.9%)
- Endometrial Carcinoma (EC) (24 samples – 4.37%)

Below are the steps followed for the data preprocessing pipeline:

- Handling missing values: The dataset had missing values across multiple features. Numerical features like Diagnosis Age and Mutation Count were imputed using the median. Categorical features like Race Category and Pregnancies Full Term Count were imputed using the most frequent category.
- Categorical encoding: The target variable Cancer Type Detailed was label-encoded into three classes: EC (0), UEC (1), and USC/UPSC (2). Other categorical features were transformed using one-hot encoding, creating an additional 49 feature columns.
- Normalization: The numerical features were standardized using StandardScaler.
- Train-Test split: The data was split into a 70% training set and a 30% test set.
- Handling imbalance: The dataset was severely imbalanced. The training set was then resampled using SMOTE.

4.2 Model Selection and Training

For analyzing and predicting the target variable cancer subtype, which is categorical, the following models were explored to determine a good fitting model:

- Logistic Regression: A baseline linear model.
- Decision Tree: To capture non-linear decision boundaries.
- Random Forest: An ensemble bagging method to reduce variance.
- Naive Bayes: A probabilistic baseline.
- Neural Network (MLP): To capture complex latent patterns.

- XGBoost: An ensemble boosting method known for high performance on tabular data.

The training and evaluation were done in 3 stages:

- Baseline models: All the 6 models were trained on the original imbalanced dataset.
- SMOTE models: All the 6 models were trained on the resampled dataset.
- RFE models: RFE feature selection was applied to the Decision Tree base model to find the most important feature set. Different feature set sizes (10, 15, 20, 25, 30) were tested to find the best balance between performance and model simplicity. 25 features from Decision Tree RFE gave the best performance. The models were trained again with just these 25 important features.

Hyperparameter tuning using GridSearchCV with 5-fold cross validation was also performed for the baseline models.

5. Experimental Evaluation

5.1 Methodology

All the trained models were evaluated based on overall accuracy and class-wise accuracy. The experiments test the hypothesis that demographic and clinical features are sufficient to predict the molecular subtypes of endometrial cancer. To maintain class balance between the training and test sets, a 70–30 stratified split was used.

The independent variables include demographic features such as age, race, and ethnicity, along with clinical features such as height, weight, and history of chronic conditions like diabetes and hypertension. The dependent variable is the cancer subtype (Cancer Type Detailed).

All three training phases were compared to identifying the model that best fits the data. This comparison helps in answering the research question stated earlier in the introduction.

The Scikit-learn framework was extensively used for model building, training, and evaluation. NumPy and pandas were used for data preprocessing and cleaning. Matplotlib and seaborn were used for visualizing data distributions and model results. The imblearn library was used to apply resampling techniques. The dedicated XGBoost library was used for model building and training the XGBoost model.

5.2 Evaluation Metrics

As the target variable is categorical, evaluation metrics such as accuracy, precision, recall, F1-score, confusion matrix, and classification report were used. Since the dataset is imbalanced, additional metrics were included: F1-score (macro) to analyze the performance of each class equally, and F1-score (weighted) to assess the overall performance while accounting for class imbalance.

5.3 Results

From the baseline models, it is evident that the random forest got the highest accuracy (80%). However, accuracy alone does not fully reflect model performance because the dataset is imbalanced. When looking at the F1-score (macro), Logistic Regression (0.55) and Decision Tree (0.54) performed better than Random Forest (0.47), meaning they handled all three classes more evenly. Naïve Bayes had the lowest overall performance across both accuracy and F1-scores, showing that it struggled with the class imbalance and feature distributions.

	Model	Test Accuracy	Test F1 (Macro)	Test F1 (Weighted)
0	Logistic Regression	0.793939	0.554649	0.796182
1	Decision Tree	0.769697	0.547416	0.761749
2	Random Forest	0.800000	0.474766	0.773919
3	Naive Bayes	0.387879	0.322661	0.490471
4	Neural Network	0.769697	0.480449	0.769697
5	XGBoost	0.775758	0.475466	0.764647

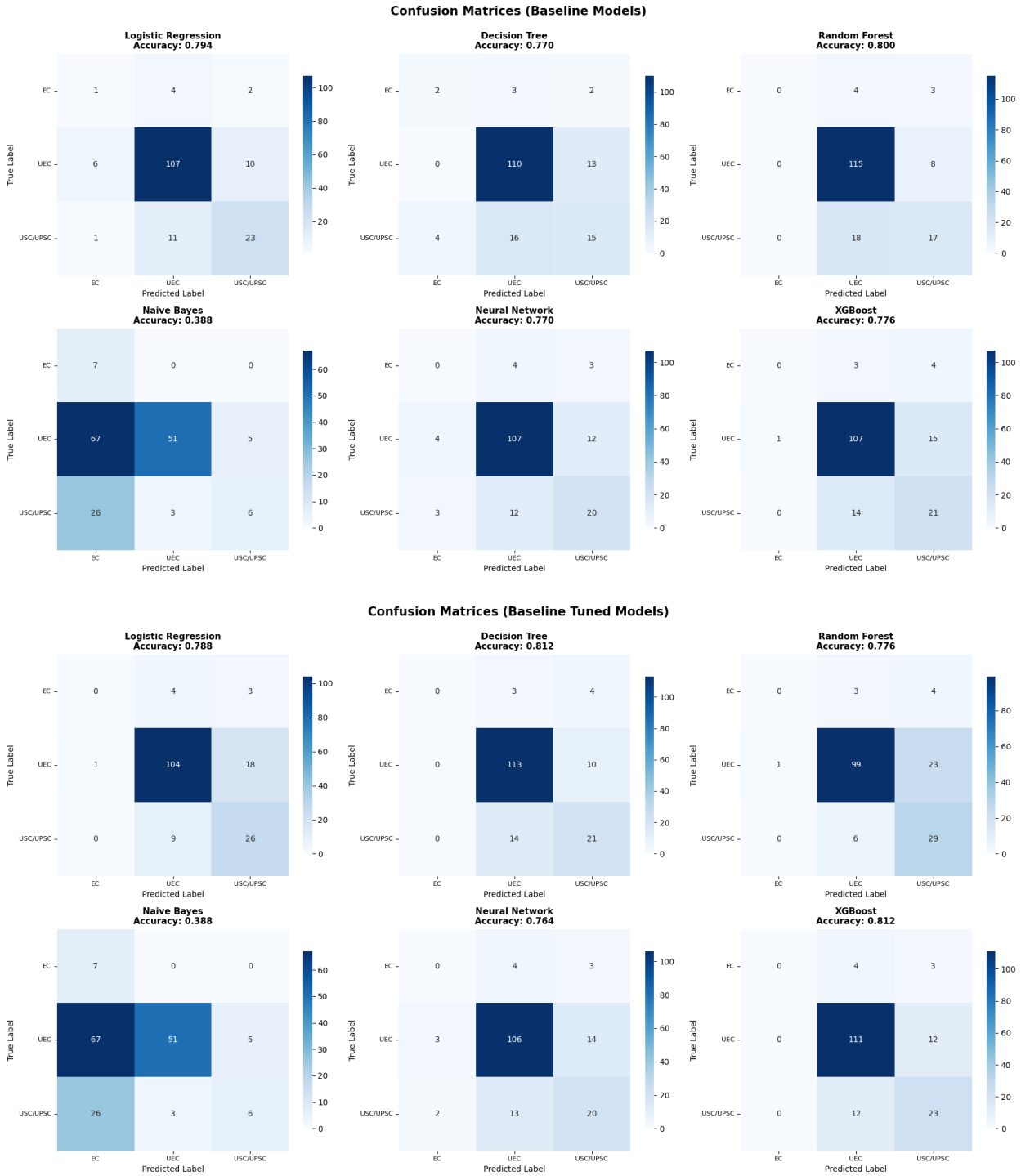
From the SMOTE models, we can see that the accuracy and F1-score (macro) improved only slightly for Random Forest and XGBoost. However, the performance of the other models either worsened or stayed about the same. This suggests that applying SMOTE did not have a positive impact on the overall model performance. The synthetic samples created by SMOTE may not have captured the true structure of the minority class, which could explain why most models did not benefit from the resampling.

	Model	Test Accuracy	Test F1 (Macro)	Test F1 (Weighted)
0	Logistic Regression	0.709091	0.545959	0.756274
1	Decision Tree	0.696970	0.487640	0.705943
2	Random Forest	0.824242	0.530854	0.813849
3	Naive Bayes	0.381818	0.314213	0.486178
4	Neural Network	0.763636	0.483341	0.767201
5	XGBoost	0.800000	0.573926	0.796581

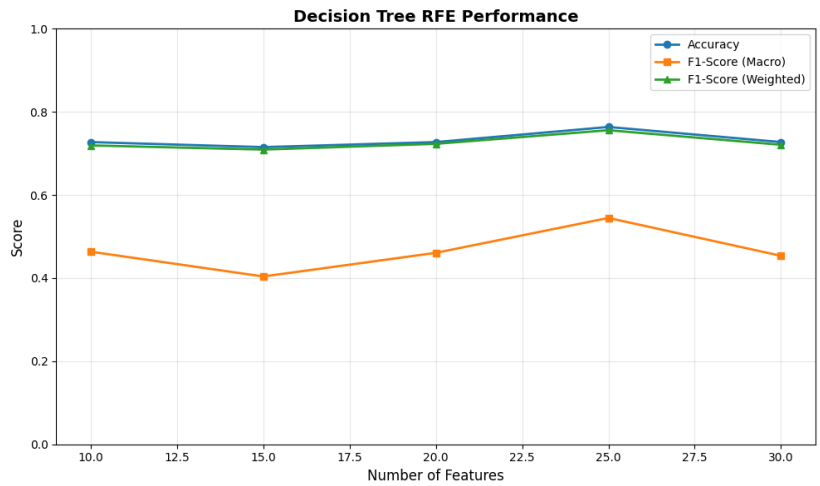
Hyperparameter tuning using GridSearchCV was then applied to the baseline models, optimizing for the F1-macro score to better handle the class imbalance. From the results, we can see that the overall performance of the Decision Tree and XGBoost models improved compared to their baseline versions. However, the improvements were modest, and the tuned models did not consistently outperform the simpler default configurations across all metrics.

	Model	Best CV Score	Test Accuracy	Test F1 (Macro)	Test F1 (Weighted)
0	Logistic Regression	0.517487	0.787879	0.500271	0.780576
1	Decision Tree	0.531174	0.812121	0.497760	0.793173
2	Random Forest	0.583113	0.775758	0.498168	0.774159
3	Naive Bayes	0.311959	0.387879	0.322661	0.490471
4	Neural Network	0.495273	0.763636	0.472448	0.760269
5	XGBoost	0.523976	0.812121	0.506046	0.795629

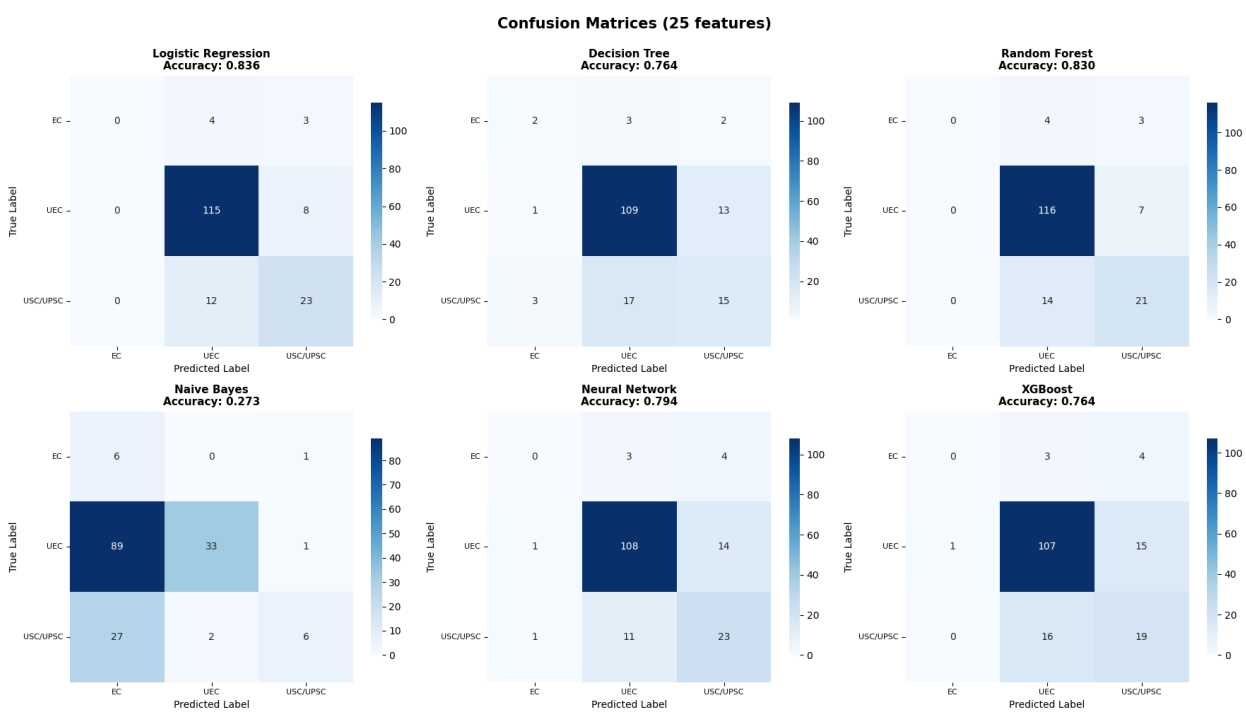
Taking a closer look at the confusion matrices for the default baseline models and the tuned baseline models shows that the default Decision Tree model was able to correctly classify 2 out of the 7 samples from the minority class. This indicates that the tuning process did not improve the model's ability to recognize the minority class, and in this case, the default configuration actually performed slightly better for minority-class detection.



RFE feature selection was then applied using the Decision Tree model to identify the most important features contributing to the prediction. From the plot below, we can see that the accuracy is highest when using 25 features. Adding more features beyond this point does not improve performance and may introduce noise, which can reduce the model's predictive power.



The baseline models were then retrained using the selected 25 features. We can observe that, with only 25 features, the accuracy remains almost the same as the full baseline models. Although Logistic Regression achieved high overall accuracy, it still failed to classify any samples from the minority class. Considering both accuracy and minority-class detection, the Decision Tree model provides the best overall fit for this dataset.



5.4 Discussion

From the experiments, it is evident that clinical features are useful for predicting molecular subtypes. The results partially support the hypothesis, as most models were able to learn meaningful patterns from the available clinical and demographic variables. Even though Naïve Bayes trained on SMOTE data correctly classified all seven minority samples, it performed poorly on the majority classes, making it unreliable overall. The other models also struggled with minority-class prediction, which shows that class imbalance remains a challenge even after resampling.

The dimensionality reduction from 49 features to 25 features maintained almost the same performance across models, which indicates that the most important clinical variables carry most of the predictive signals. This also highlights the strength of simpler and more interpretable models, such as Decision Trees, which handled the reduced feature set effectively. Overall, the results show that clinical features have predictive value, but improving minority-class detection will require either more data or additional feature types.

6. LIMITATIONS

The main assumptions made for this project are the imputations for missing data which may have introduced bias. Also, training the models with additional data on serological markers could potentially improve performance. Another limitation is that the dataset size is relatively small which may not be sufficient to train complex models like neural networks. With a larger dataset, a different model might fit the data better and produce even stronger results.

7. CONCLUSIONS AND FUTURE WORK

Across all the experiments, baseline decision trees with just 25 important features performed better on the data. This shows that simpler models can perform just well or even better than complex models on this type of structured clinical data. This can further be enhanced by including serological markers data. The insights obtained from this project show that it is possible to build lightweight, practical models for clinical decision support systems especially in low resource constrained healthcare settings

8. DATA AVAILABILITY

The dataset is available at cBioPortal for Cancer Genomics. This dataset is shared under the TCGA Data Usage Policy. I have uploaded the dataset and code in the Moodle submission link.

9. ACKNOWLEDGEMENTS

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10. REFERENCES

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